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Basics of Measurement Invariance

Table of contents

Overview 1
Statistical and Causal Perspectives 1
Data and preprocessing 2
Model specification 3
Multiple-group CFA 4
Illustration of invariance assumptions 4
Model comparison 6
Interpretation 7
References 7

```
library(lavaan)
library(lavaangui)
library(semTools)
library(modelsummary)
library(tinytable)
library(dplyr)
library(ggdag)
library(dagitty)
```

Overview

This note demonstrates measurement invariance testing of the Self-Oriented Perfectionism (SOP) scale across gender using multiple-group CFA. We follow the standard sequence of nested models: configural (same pattern of fixed/free loadings), metric (equal loadings), scalar (equal loadings + intercepts), and strict (add equal residual variances/residual covariances). Data and exclusion criteria follow Williams & Edwards (2021; OSF: https://osf.io/m5dy6/).

Statistical and Causal Perspectives

To put it simply, measurement invariance is the idea that some attributes are measured in the same way across different groups, times, or contexts. This is a fundamental assumption in quantitative science. One would not be able to say A is taller than B if the ruler used to measure A is different from the one used to measure B. Similarly, in psychological measurement, we would not be able to say that A has better well-being than B, if the mental "ruler" is different for A and B. Measurement invariance testing is a way to check whether the "ruler" is the same across groups.

The complication in psychology is that we are measuring abstract constructs, and our instruments (e.g., questionnaires) are imperfect indicators of those constructs. Imperfect measurement means that there are measurement errors. So instead of asking whether the "ruler" is the same across groups, we ask whether the "ruler" gives equivalent distributions of scores across groups. So, from a statistical perspective, for a grouping variable G and an observed score Y, measurement invariance means

$$P(Y \mid G = A) = P(Y \mid G = B) = \ldots$$

for all groups.

```
dag <- dagitty::dagitty("dag { group -> eta; eta -> Y; group -> Y }")
ggdag::ggdag(dag, text = TRUE) +
    ggplot2::labs(x = NULL, y = NULL) +
    ggdag::theme_dag()
```

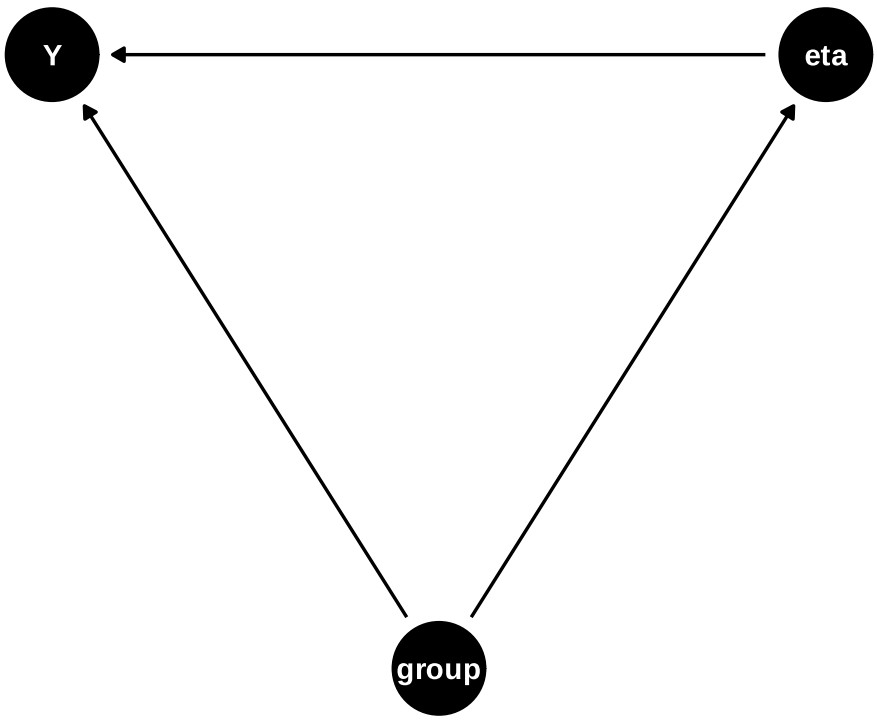

Figure 1: Causal diagram: grouping -> latent construct (eta) -> observed score (Y); grouping also directly influences Y.

Data and preprocessing

```
we_path <- here::here("data", "we_data.csv")
if (!file.exists(we_path)) {
    download.file("https://osf.io/download/6mdb4/", we_path)
}
```

```
we_raw <- read.csv(we_path)
we_raw <- dplyr::rename(we_raw, Att_check = SOP_6)
# Exclude: under-18, non-students, >6 missing study items, failed attention
check, duplicates
we_vars <- c(
    paste0("PASS_", c(1:9, 16:18)),
    paste0("NGSE_", 1:8),
    paste0("SOP_",  1:5)
)
we_raw <- we_raw[is.na(we_raw$Age) | we_raw$Age != 1, ]
we_raw <- we_raw[is.na(we_raw$Student) | we_raw$Student == 1, ]
we_raw$nmiss <- rowSums(is.na(we_raw[, we_vars]))
we_raw <- we_raw[we_raw$nmiss <= 6, ]
we_raw <- we_raw[!is.na(we_raw$Att_check) & we_raw$Att_check == 5, ]
we_raw <- we_raw[!duplicated(we_raw$PID_deID), ]
```

```
sop_data <- we_raw[, paste0("SOP_", 1:5)]
nrow(sop_data)
```

```
[1] 575
```

The analysis uses the SOP items and a derived `gender` grouping variable (male, female). The R script `notes/R/invariance.R` performs data preparation and applies the preregistered exclusions; we reuse those objects here.

```
sop_data_mg <- cbind(
    sop_data,
    gender = factor(we_raw$Gender, levels = 1:3,
                    labels = c("male", "female", "nonbinary"))
)
# Only use males and females as sample size is small for nonbinary group
sop_data_mg <- sop_data_mg[sop_data_mg$gender %in% c("male", "female"), ]
```

Model specification

The model and estimation code are sourced from `notes/R/invariance.R` via chunk labels; see the `configural` chunk for the model definition.

```
model_1f <- "
  SOP =~ SOP_1 + SOP_2 + SOP_3 + SOP_4 + SOP_5
  SOP_1 ~~ SOP_2
"
fit_config <- cfa(
    model_1f,
    data = sop_data_mg,
    std.lv = TRUE,
```

```
    estimator = "MLR",
    missing = "FIML",
    group = "gender"
)
```

Multiple-group CFA

We estimate the nested models (configural → metric → scalar → strict) using robust maximum likelihood (`MLR`) and FIML for missing data. The code chunks below are populated from `notes/R/invariance.R`.

```
fit_metric <- cfa(
    model_1f,
    data = sop_data_mg,
    std.lv = TRUE,
    estimator = "MLR",
    missing = "FIML",
    group = "gender",
    group.equal = "loadings"
)
```

```
fit_scalar <- cfa(
    model_1f,
    data = sop_data_mg,
    std.lv = TRUE,
    estimator = "MLR",
    missing = "FIML",
    group = "gender",
    group.equal = c("loadings", "intercepts")
)
```

```
fit_strict <- cfa(
    model_1f,
    data = sop_data_mg,
    std.lv = TRUE,
    estimator = "MLR",
    missing = "FIML",
    group = "gender",
    group.equal = c("loadings", "intercepts", "residuals",
                    "residual.covariances")
)
```

Illustration of invariance assumptions

```
library(ggplot2)
set.seed(2026)
```

```
n <- 200
types <- c("Configural", "Metric", "Scalar", "Strict")
err <- rnorm(n, mean = 0, sd = 1)
params <- list(
     # non-strict models: groups have different residual SDs and different
intercepts (means)
    Configural = list(
        A = list(int = 0, xmean = 0, xsd = 1, slope = 0.8, sd = 0.6),
        B = list(int = -0.5, xmean = 0.6, xsd = 1.2, slope = 0.5, sd = 0.9)
    ),
    Metric = list(
        A = list(int = 0, xmean = 0, xsd = 1, slope = 0.8, sd = 0.6),
        B = list(int = -0.5, xmean = 0.6, xsd = 1.2, slope = 0.8, sd = 0.9)
    ),
    # show scalar with equal slopes but different latent means to illustrate
mean differences
    Scalar = list(
        A = list(int = 0, xmean = 0, xsd = 1, slope = 0.8, sd = 0.9),
        B = list(int = 0, xmean = 0.6, xsd = 1.2, slope = 0.8, sd = 0.6)
    ),
   # strict invariance: residual SDs and latent means are equal across groups
    Strict = list(
        A = list(int = 0, xmean = 0, xsd = 1, slope = 0.8, sd = 0.7),
        B = list(int = 0, xmean = 0.6, xsd = 1.2, slope = 0.8, sd = 0.7)
    )
)
df <- do.call(
    rbind,
    lapply(types, function(t) {
        p <- params[[t]]
        do.call(
            rbind,
            lapply(c("A", "B"), function(g) {
                par <- p[[g]]
                x <- rnorm(n, mean = par$xmean, sd = par$xsd)
                y <- par$int + par$slope * x + err * par$sd
                data.frame(type = t, group = g, x = x, y = y)
            })
        )
    })
)

df$group <- factor(df$group, labels = c("Group 1", "Group 2"))

p <- ggplot(df, aes(x = x, y = y, color = group, fill = group)) +
    stat_ellipse(geom = "polygon", level = 0.68, alpha = 0.25, color = NA) +
    geom_smooth(
        method = "lm",
```

```
        se = FALSE,
        linewidth = 0.9,
        show.legend = FALSE
    ) +
    facet_wrap(~type, nrow = 2) +
    labs(
        x = "Latent trait (z)",
        y = "Item score (observed)",
        color = "Group",
        fill = "Group"
    ) +
    theme_minimal()
p
```

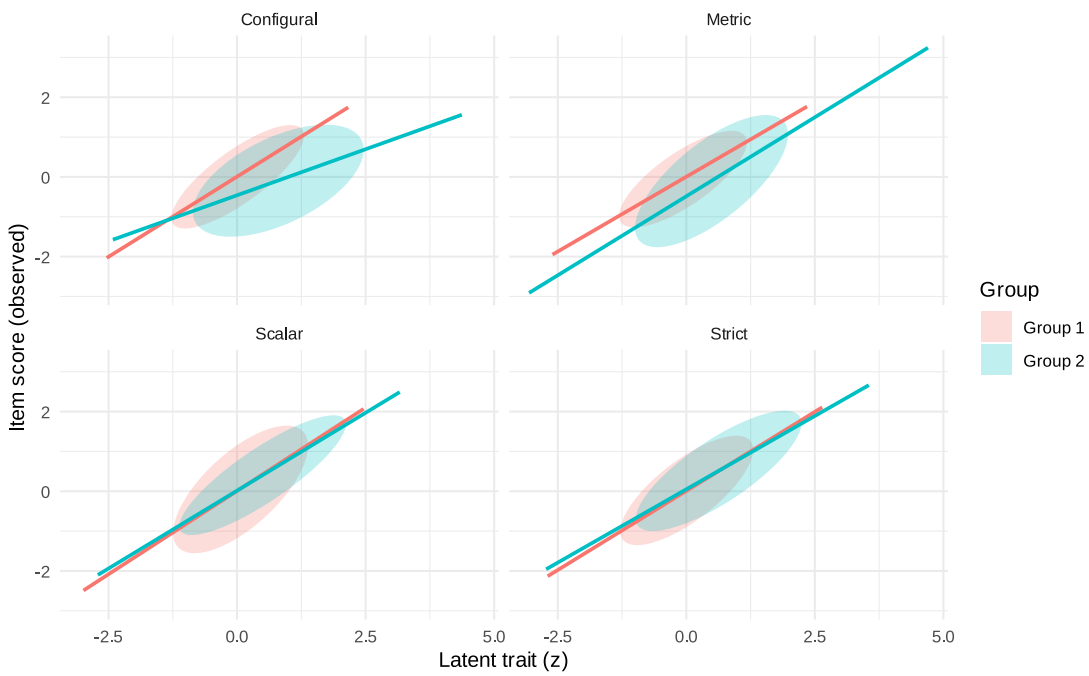

Figure 2: Illustration of invariance assumptions: latent (x) vs item score (y).

Model comparison

We compare nested models using a robust likelihood ratio test.

```
lavTestLRT(fit_config, fit_metric, fit_scalar, fit_strict)
```

```
Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

lavaan->lavTestLRT():
   lavaan NOTE: The "Chisq" column contains standard test statistics, not the
   robust test that should be reported per model. A robust difference test is
```

```
   a function of two standard (not robust) statistics.

           Df    AIC    BIC  Chisq Chisq diff Df diff Pr(>Chisq)
fit_config  8 7172.5 7311.4 23.797
fit_metric 12 7166.0 7287.5 25.265     1.7679       4     0.7783
fit_scalar 16 7162.8 7267.0 30.092     4.7616       4     0.3126
fit_strict 22 7155.2 7233.3 34.466     3.3041       6     0.7698
```

Interpretation

- Configural invariance indicates the same factor structure holds across groups (pattern of fixed/free loadings).
- Metric invariance (equal loadings) supports comparing covariances and regression coefficients across groups.
- Scalar invariance (equal intercepts) is required to compare latent means across groups.
- Strict invariance additionally constrains residual variances and residual covariances.

If the nested test indicates misfit when imposing equality constraints, evaluate practical fit changes (CFI, RMSEA) and inspect modification indices to identify noninvariant parameters. Partial invariance (freeing a small set of parameters) is commonly investigated when full invariance is untenable.

References

Williams, M. N., & Edwards, S. R. (2021). Conceptual replication of Seo (2008). OSF: https://osf.io/m5dy6/